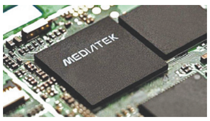
NEW PRODUCTS

MAKE IN INDIA

COMPONENTS & CHIPS

SoC for 5G devices

MediaTek has unveiled the Dimensity 6100+ chipset, designed to empower the next generation of mainstream 5G devices with better performance and higher power efficiency. The chipset offers a range of impressive features at a competitive price point, making budget smartphones more powerful and feature-rich. The SoC offers an enhanced 5G modem supporting up to 140MHz 2CC 5G carrier aggregation, delivering reliable connectivity and improved battery life for a longer user experience. The Dimensity 6100+ chipset is equipped with two Arm Cortex-A76 big cores and six Arm Cortex-A55 efficiency cores, ensuring powerful performance. It supports AI-powered cameras, enabling features like AI-bokeh for stunning portraits and selfies, and AI-colour technology for immersive visual experiences. With support for up to 108MP Non-ZSL camera, 2K 30fps video capture, and a 10-bit display, the Dimensity 6100+ chipset will allow smartphone manufacturers to offer better smartphone performance at a cost-effective price.



MediaTek
https://www.mediatek.com

Wireless power charging SoC

Indie Semiconductor has introduced the iND87200, an integrated automotive wireless power charging system-on-chip (SoC) that revolutionises in-cabin portable device charging designs. Designed to comply with the wireless power charging (WPC) 'Qi' standard, the iND87200 simplifies and accelerates the development of cost-effective in-cabin device charging solutions. The SoC supports the Qi 2.0 standard, ensuring enhanced charging reliability even while the vehicle is in motion. The iND87200 features a dual-core design with an Arm Cortex-M4F processor, 2MB of embedded Flash, and 256kB of SRAM, along with a dedicated Arm Cortex-M0 processor for the WPC stack. This architecture allows for the separation of compute resources, enabling the execution of user-specific software without timing or interrupt constraints. Indie Semiconductor's wireless charging solution integrates essential components such as power management, DC-DC converter, signal conditioning, WPC inverter drivers, power FETs, LED drivers, and fan drivers. It supports various serial interfaces, providing multiple connectivity options for the vehicle and peripherals. The iND87200 reduces the external bill-of-materials by almost half compared to existing discrete implementations, simplifying system complexity, improving reliability, and enabling faster time-to-market for automotive OEMs.

Indie Semiconductor
www.indiesemi.com

Advanced data-logging F-RAM devices

Infineon Technologies has launched two advanced Ferroelectric RAM (F-RAM) memory devices in response to the growing demand in the automotive event data recorder (EDR) market. The 1Mbit and 4Mbit data-logging components instantly capture and store critical data long-term. The 1Mbit units are the first automotive-qualified serial F-RAMs in the industry. They function effectively between -40°C and +125°C. The devices offer fast, reliable read/write speeds of up to 50MHz in serial peripheral interface (SPI) and 108MHz in quad serial peripheral interface (QSPI) mode. They have a 10 trillion read/write cycle endurance, suitable for logging data every 10 microseconds for over 20 years. Their zero delay write captures data until just before a trigger event. Operating between 1.8V and 3.6V, they come in an 8-pin small outline integrated circuit (SOIC) package. Notably, they retain data for 100 years post power loss, ensuring data reliability for many applications.

Infineon Technologies
https://www.infineon.com

IR sensor for motion detection in building automation

STMicroelectronics has launched a human-presence and motion detector. The device is designed to augment security systems, home automation equipment, and IoT devices, which generally utilise passive infrared (PIR) sensing technology. The STHS34PF80 sensor, equipped with thermal transistors, can detect stationary objects. Whereas conventional PIRs rely on a Fresnel lens to sense